1. $(1+x)ydx+(1-y)xdy=0$ bu o'zgaruvchilari ajraladigan differensial tenglamadir.

Yechish. O'zgaruvchilarni ajratish uchun tenglamaning har bir hadini $xy\neq 0$ ga bo'lamiz.

$\frac{1+x}{x}dx+\frac{1-y}{y}dy=0$, integrallaymiz $ln|x|+x+ln\,y-y=ln\,C$,

$ln\left|\frac{xy}{C}\right|=y-x$; $\frac{xy}{C}=e^{y-x}$, $xy=Ce^{y-x}$ bu tenglamaning umumiy yechimi. ▶

O'zgaruvchilari ajraladigan differensial tenglamalarning umumiy yechimlarini toping

2. $x^2\cdot y^1=(1-x^3)/y;$

3. $y\cdot y^1=(1+2x)/y^2;$

4. $\frac{dy}{dx}=\frac{2x^2y}{x^3-y^3}$ tenglama bir jinsli differensial tenglama.

Yechish. Tenglamani $y=ux$ almashtirish bajarib integrallaymiz,

$y=ux$ dan $\frac{dy}{dx}=u+\frac{du}{dx}x;$

$$u+x\frac{du}{dx}=\frac{x^2ux}{x^3-u^3x^3};\quad x\frac{du}{dx}=\frac{u}{1-u^3}-u;\quad x\frac{du}{dx}=\frac{u^4}{1-u^3};$$

o'zgaruvchilarni ajratamiz

$\frac{dx}{x}=\frac{1-u^3}{u^4}du$ integrallaymiz $ln\,x+ln\,C=\frac{1}{3u^3}-ln\,u$;

$$ln\,Cxu=\frac{-1}{3u^3};\qquad ln\,Cx\frac{y}{x}=-\frac{x^3}{3y^3};\qquad ln\,Cy=-\frac{x^3}{3y^3}$$

tenglamaning umumiy yechimi. ▶

? Quyidagi egri chiziqlar oilalarining differensial tenglamalarini tuzing

5. $x^2 + y^2 = cx;$

6. $y^2 = 2cx;$

7. $y = c_1 \cdot e^x + c_1 \cdot e^{-x};$

8. $y = (c_1 + c_2 x) \cdot e^{-2x}.$

? O'zgaruvchilari ajraladigan differensial tenglamalarning umumiy yechimlarini toping.

9. $x \cdot dx / y = \sqrt{(x^2 + 1)} \cdot dy;$

10. $y^1 \cdot tgx - y = 10;$

11. $\sqrt{(1 - x^2)} \cdot dy + x \cdot \sqrt{(1 - y^2)} \cdot dx = 0;$

12. $y^1 = e^{x+y}$

? Koshi masalalarini yeching:

13. $(1 + x^2) \cdot y^1 = 1 + y^2, \quad y_{x=0} = 1;$

14. $y^1 = tgx \cdot ctgy, \; y_{x=0} = \pi/4;$

15. $(1 + e^{2x}) \cdot y^2 \cdot y^1 = e^x, \quad y_{x=0} = 0;$

16. $y \cdot \ln y \cdot dx = \sin x \cdot dy, \quad y_{x=\pi/2} = 1$

? Bir jinsli differensial tenglamalarni yeching:

17. $(x - y) \cdot dy - y \cdot dx = 0;$

18. $y^1 = y^2 / x^2 - 1;$

19. $(x^2 - y^2) \cdot dy = 2xy \cdot dx;$

20. $y^1 = y/x + x/y;$

21. $y^1 = 2xy/(3x^2 - y^2), \quad y_{x=0} = 1;$

22. $(y - x \cdot y^1) \cdot arctgy/x + x = 0, \quad y_{x=1} = 0.$

? Chiziqli differensial tenglamalarni yeching:

23. $y' + y = 2x;$

24. $y^1 + 4y/x = x^3;$

25. $y^1 + 2xy = x \cdot e^{-x2};$

26. $y^1 + y = \sin x;$

27. $y^1 + 2y = e^x;$

28. $y^1 - tgx \cdot y = 1/\cos x; \quad y_{x=0} = 0$

29. $y^1 - y/(x^2 + x) = 1; \quad y_{x=1} = 0.$

? Bernulli tenglamalari umumiy yechimlarini toping:

30. $y^1 + y/x = -x \cdot y^2;$

31. $y^1 + y/(x+1) = -y^2;$

32. $(x^2/y - y^3) \cdot dy - x \cdot dx = 0;$

33. $y^1 - tgx \cdot y = -\cos x \cdot y^2;$

34. $y^1 - 4/x = x \cdot \sqrt{y} = 0.$

? Iqtisodiy mazmundagi masalalarni yeching:

35. Raqobat sharoitida biror bir tovarga bo'lgan talab va taklif funksiyalari quyidagicha bo'lsin

$$Q_d = 35 - 0.5p \quad \text{va} \quad Q_s = -4 + 0.8p.$$

Agar narx talab va taklif orasidagi muvozanat shartini qanoatlantirmasa, u holda bozordagi raqobat ta'sirida u quyidagicha o'zgaradi

$$\frac{dp}{dt} = 0,5(Q_d - Q_s).$$

IKKINCHI TARTIBLI DIFFERENSIAL TENGLAMALAR

1. $y'' - 8y' + 7y = 0$ tenglama umumiy yechimini quring.

Yechish. Xarakteristik tenglama $\lambda^2 - 8\lambda + 7 = 0$ ko`rinishga ega va uning ildizlari $\lambda_1 = 1, \lambda_2 = 7$. Natijada, chiziqli erkli $y_1 = e^x$; $y_2 = e^{7x}$ yechimlarni olamiz. Tenglamaning umumiy yechimi:

$$y = C_1 e^x + C_2 e^{7x}. \quad \blacktriangleright$$

O`zgarmas koeffitsientli chiziqli bir jinsli differensial tenglamalarning umumiy yechimini toping.

2. $y'' + 2y' + y = 0;$

3. $y'' + 6y' + 25y = 0;$

4. $y'' - 4y' = 0.$

5. $y'' - 4y' + 5y = 12\sin x + 4\cos x$ tenglamaning xususiy yechimini toping.

Yechish. Xarakteristik tenglamani yechamiz.

$$\lambda^2 - 4\lambda + 5 = 0,$$

$\lambda = 2 \pm i$. Bizning holatda $\alpha = 0$ va $\beta = 1$ bo'lib, xarakteristik tenglamaning ildizi emas. Demak, xususiy yechim quyidagicha qidiriladi

$$y = A\sin x + B\cos x.$$

Funksiya hosilalarini aniqlaymiz:

$$y' = A\cos x - B\sin x$$

$$y'' = -A\sin x - B\cos x.$$

y, y', y'' ifodalarni tenglamaga qo`yamiz va soddalashtiramiz

$$(-A\sin x - B\cos x) - 4(A\cos x - B\sin x) + 5(A\sin x + B\cos x)$$

$$=12\sin x+4\cos x \text{ yoki}$$

$$(4A+4B)\sin x+(4B-4A)\cos x=12\sin x+4\cos x.$$

Bundan

$$\begin{cases}4A+4B=12,\\4B-4A=4,\end{cases}$$

yoki $A=1,\ B=2$. Xususiy yechim

$$y_{xususiy}=\sin x+2\cos x.$$

Demak, umumiy yechim

$$y=e^{2x}(C_1\sin x+C_2\cos x)+\sin x+2\cos x,$$

bu yerda C_1 va C_2 ixtiyoriy o'zgarmas sonlar ▶

? O`zgarmas koeffitsientli chiziqli bir jinsli differensial tenglamalarning umumiy yechimini toping.

6. $a)\ y''-y=0;\ b)\ 4y''+8y'-5y=0;\ c)\ y''-6y+10y=0$

7. $a)\ y''+10y'=0;\ b)\ y''-6y'+8y=0;\ c)\ 4y''+4y'+y=0$

8. $a)\ y''+25y=0;\ b)\ y''+6y'+9y=0;\ c)\ y''+2y'+2y=0$

9. $a)\ y''+5y=0;\ b)\ 9y''-6y'+y=0;\ c)\ y''+6y'+8y=0$

10. $a)\ 6y''+7y'-3y=0;\ b)\ y''+16y=0;\ c)\ 4y''-4y'+y=0$

11. $a)\ y''-6y'+13y=0;\ b)\ y''-2y'-15y=0;\ c)\ y''-8y'=0$

12. $a)\ y''+6y'+10y=0;\ b)\ y''-4y'+4y=0;\ c)\ y''-5y'+4y=0$

? O`zgarmas koeffitsientli chiziqli bir jinsli bo`lmagan differensial tenglamalarning umumiy yechimini toping.

13. $y''+2y'+37y=37x^2-33x+74$

14. $y''+4y'=15e^x$

15. $y''-4y'+5y=(24\sin x+8\cos x)e^{-2x}$

16. $y''+3y'-y=11\cos x-7\sin x$

17. $y''-4y'+29y=104\sin 5x$

18. $y''+4y'+20y=4\cos 4x-52\sin 4x$

19. $y''+2y'-24y=6\cos 3x-33\sin 3x$

20. $y''+16y=80e^{2x}$

21. $y''+y'=-4\cos x-2\sin x$

22. $y''+4y'+29y=26e^{-x}$

23. $y''+8y'+25y=18e^{5x}$

24. $3y''-5y'-2y=6\cos 2x+38\sin 2x$

25. $y''+5y'=72e^{2x}$

26. $y''-4y'=8-16x$

27. $y''-3y'+2y=3\cos x+19\sin x$

? Iqtisodiy mazmundagi masalalarni yeching:

28. Tovar va materiallar zahirasi hisobga olinadigan narx moslashuvchanligi $\ddot{p}=\alpha(\dot{Q}_d-\dot{Q}_s)-\beta\left[Q_d-Q_s\right]$ modelida $\alpha=0{,}3$, $\beta=0{,}1$, $Q_d=35-0.5p$ va $Q_s=-4+0.8p$ bo'lsin. Agar $p(0)=500$, $\dot{p}(0)=2$ narx dinamikasini ifodolovchi $p(t)$ funksiyani toping.

29. Valrasa narx moslashuvchanligi $\ddot{p}=\alpha\left(B\dot{p}-m\dot{N}\right)$, $\dot{N}=\gamma(p-\overline{c})$ modelida $Q_d=40-0.6p$, $\alpha=0{,}3$, $\gamma=0{,}1$, $m=0{,}25$, $\overline{c}=500$, $p(0)=400$, $\dot{p}(0)=1{,}2$ bo'lsa, narx dinamikasini ifodolovchi $p(t)$ funksiyani toping.